

Financial Management and Valuation

Mini Cases Unit 5

SBWL Finance

Winter Term 2026

Please upload your solution and indicate which Mini-Cases (MCs) you have solved before **January 19, 2026 at 10.00pm**. To do so, go to the assignment section on Canvas and choose “Unit 5 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). There will be no deadline extensions! You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Exercise 5.1

The car manufacturer Motoren Werke Brigittenau (MWB) trades at €90 with a historical beta of 1.23. Analysts forecast next years earnings at €10.6 and next years share price at €96 and a constant growth rate. The risk-free rate is 2% p.a. and we assume that CAPM holds.

a) What is the expected return on the market portfolio based on the given information?

After long negotiations, MWB managed to acquire one of its biggest competitors, Gaudi. After the acquisition, the beta increases to 1.6. The dividends are now expected to be €15. All other variables and the percentage by which the share price is expected to grow are unaffected by the change.

b) What is the new share price?

Exercise 5.2

Consider an investment opportunity with a lifetime of four years. The project is expected to pay €50,000 at the end of the first year. Cash flows then increase by 6% annually for the next two years. At the end of the fourth year, the project incurs a terminal cash outflow of €−30,000. Assume an expected market return of 10% per year, a risk-free rate of 3% per year, and a project beta of 0.9. Determine the present value of the project.

Exercise 5.3

Analyze the following investment over the course of three years with a multi-factor model. Assume that the betas are constant.

Factor Exposure	Beta
Market Risk	0.85
Size	-0.12
Value	0.34
Momentum	0.27

Factor Risk Premium in %	Year 1	Year 2	Year 3
Market Risk	5.2	6.1	4.8
Size	1.3	0.9	0.4
Value	2.4	2.9	3.1
Momentum	3.6	4.1	3.7

The risk-free rate is 2 percent for year 1, 4 percent for year 2, and 3 percent for year 3. In year 1 the investment will produce a cash flow of 20 million, in year 2 the cash flow is expected to be 30 million, and in year 3 the cash flow is either 50 million with a probability of 45 percent or 20 million with a probability of 55 percent. What is the present value of the investment?

Exercise 5.4

Zweite Bank Group is not only one of the largest banks but also among the largest companies of the country Laustria, currently trading at €90. This year they managed to get €7.4 earnings per share (EPS). After larger successes following the expansion to Soland the EPS have been predicted to be €9.2. The same analysts also forecast major adjustments of goodwill and off-market securities leading to expenses of 12m Euro. There are 410,514,384 shares outstanding. Zweite Bank Group's beta is at 0.89. The risk-free rate is 0.02 and the market returns are expected to be 0.05.

Assuming that the CAPM holds, what is the expected growth rate of Zweite Bank Group?

Exercise 5.5 You are an analyst for a private equity fund. Currently, the fund wants to expand into the field of food and beverages, and you are tasked to join their due diligence analysis. Through connections of the fund owner with a university buddy, your team was asked to look into Laze Drinks.

The table below presents information about the earnings forecast for the firm and the distribution of earnings across the four business lines:

Expected earnings in EUR	800,000
Fraction of expected earnings from energy drinks	0.40
Fraction of expected earnings from beer	0.25
Fraction of expected earnings from bottled water	0.15
Fraction of expected earnings from tea	0.20

Further, you are given a table providing information about the pure player competitors.

(in EUR)	Jonster Energy	Jottakringer	Jöslauer	Jack Jabel
Industry	Energy drink	Beer	Bottled water	Tea
Price per share	60	57	41	30
Earnings per share	1.1	1.3	1.2	0.6

You are asked to find answers to the following questions.

- What are the price-earnings ratios of each of the competitors?
- What are earnings can we expect from Laze Drinks lines of business?
- In a first meeting the management of Laze Drinks wanted a valuation of their company of 50 million Euros. Based on a P/E ratio comparison, is this a fair price?
Hint: Assume stable price-earnings ratios across industries.

Exercise 5.6

Following the latest increase in gold prices, the management of a mining company is in negotiations with the government of a country in the global south to build a gold mine. Things look promising, so that construction the mine could start by the end of Q1 this year. The building of the mine costs 100 million USD. Additionally, the firm has to adhere to the countries environmental, social and governance (ESG) standards. The required health and safety investments sum up to 30 million USD and CO2 emission standards also require the purchase of special machinery and air filters that sum up to 50 million USD. All cost occur at the beginning of 2026. The mine will be operational at the beginning of

2027. In the first year the mine will be 80 percent operational extracting 500,000 tons of gold ores. In the second year the mine will be fully operational. The mine can be operated for 6 years.

There are elections by the end of 2026 and it is clear already that the opposition will be governing the country next year. This opposing party has promised to reduce ESG regulations. So, alternatively the mining company could also wait with the construction start in 2027. In this case the base costs would be the same but the costs for required health and safety investments are reduced to 10 million USD and the CO2 emission reduction investments drop to 15 million USD. Again, it takes 1 year to construct the mine. In the first year of operation the mine will be 80 percent operational extracting 500,000 tons of gold ores. In the second year the mine will be fully operational. Due to the worse equipment the mine can only be operated for 5 years. In both cases the company can extract 1.5 grams of gold from one ton of gold ore. The beta for the project is 2.9. The market is expected to return 8 percent next year and 7 percent after that. The risk-free rate is 4 percent. The mining company is risk neutral. The gold price is at 130,000,000 USD/ton and is going to be constant in expectations over the next ten years. For simplicity assume that there are no costs involved in the process apart from the initial investments. Also assume that the management has to decide now if they build the mine this year or wait for another year. The management has to make a decision now and cannot change it afterwards.

- (a) What is the NPV of the project without any delays?
- (b) What is the value of the option to delay? Should the company choose to delay (neglect ethical concerns)?